

STEP 10-V — DISPARITY CORRELATION VERIFICATION (SE vs SHADOW SHARE)

Scope

Verification only. Four correlation figures were computed on the supervision side during the disparity reframe and are now stated in the paper (§2.7 rationale and §4.4) marked pending verification. Your task is to reproduce all four **independently** from the audited data files and confirm they match, or flag any drift. Do not edit any paper file; do not recompute the disparity operators themselves — only these correlations.

Values to reproduce (currently in the paper)

Zero-order, SWIID Gini standard error vs shadow-economy share, across all 20 countries: Spearman $\rho = 0.828$; Pearson $r = 0.924$. **Partial**, same pair controlling for the UNDP income sub-index: $r = 0.413$ ($p = 0.071$) on values, and $r = 0.552$ ($p = 0.012$) on ranks. All four appear in both §2.7 and §4.4; a single verification covers every occurrence.

Tasks

#	Task	Standard
1	Assemble the three columns by country from the audited files: Gini SE from Table_5B_Gini_SE.csv (column "Gini SE (raw)"), shadow-economy share from Table_5A_Shadow_Econ.csv (column "Shadow Econ (I)"), and the income sub-index from Table_4_HDI_FN_Preview.csv (column "I_income_adj (approx.)"). Merge on Country; confirm 20 rows with no missing cells.	20 countries matched; column sources named exactly; no silent drops or renames
2	Zero-order correlations. Compute Spearman ρ and Pearson r between Gini SE and shadow share. Report to three decimals with p-values.	$\rho = 0.828$, $r = 0.924$ reproduced to 3 d.p.; any deviation beyond ± 0.005 flagged with the computed value and a note on the likely cause
3	Partial correlations. Correlate Gini SE with shadow share controlling for the income sub-index — i.e. residualise each of the two variables on the income column (with intercept) by OLS, then correlate the residuals. Do this once on the raw values (expect $r = 0.413$) and once on the rank-transformed columns (expect $r = 0.552$). Report both with p-values.	Both partials reproduced to 3 d.p.; method (residual-on-residual) stated; p-values reported; deviation beyond ± 0.005 flagged
4	Deliver a short report: the merged 20-row table, the four statistics with p-values, the exact code or tool used, and one line confirming match-or-drift against the four target values above.	Merged table included; four statistics reported; explicit match/drift statement; code or tool named

Watch-outs

Partial correlation is residual-on-residual, not a three-way correlation matrix — regress SE on income and shadow on income separately, then correlate the two residual vectors; the rank version does the same after rank-transforming all three columns. **The zero-order and partial figures are meant to differ sharply** (0.924 vs 0.413): that attenuation is the expected result, not an error — do not "correct" the partial toward the zero-order. **Use the raw SE column** (Gini SE (raw)), not any normalised SE. **Report what you get** even if it differs; a flagged discrepancy is a correct outcome, a silently adjusted number is not.

Sequencing

Independent of all other open tasks; one person, one sitting. On confirmation, the §2.7 and §4.4 pending-verification status is cleared on the supervision side. If drift is reported, do not proceed to any paper change — return the discrepancy for adjudication first.